# Hilbert Spaces in Quantum Mechanics

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**Abstract**

The mathematical arena of quantum mechanics is a Hilbert space, and this article builds that space from first principles, assuming only linear algebra and the definition of a limit. The inner product supplies the geometry: orthogonality, the Cauchy–Schwarz inequality, and the orthogonal projection that measurement will turn out to be. Completeness—every Cauchy sequence of vectors converges in norm—is the property that elevates an inner product space to a Hilbert space; it fails for the continuous functions under the mean-square metric and is restored by the Lebesgue integral, which is why the infinite expansions of quantum mechanics are genuine limits rather than figures of speech. Orthonormal bases give Parseval's identity and the Riesz–Fischer theorem, which identifies the square-integrable functions $L^2$ with the sequence space $\ell^2$. Self-adjoint operators and the spectral theorem supply the language of observables. The framework is then applied: states are unit vectors, observables are self-adjoint operators, probabilities are squared inner products, and evolution is unitary. A closing section reads one state in the position and momentum bases, where a Fourier transform connects the two wave functions and the rigged Hilbert space houses Dirac's unnormalizable kets. Each idea is stated once, proved in the simplest case where the proof is transparent, and then interpreted physically.

**Keywords:** Hilbert space; inner product; completeness; orthonormal basis; self-adjoint operator; spectral theorem

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## Introduction

Quantum mechanics describes a physical system not by what it is but by the set of things that can be measured about it. The mathematical structure that organizes those possibilities is a Hilbert space. Every state is a vector; every observable an operator; every probability the squared modulus of an inner product. This article builds those three statements from the ground up.

The idea that functions could be the points of a space grew out of Hilbert's (1912) work on integral equations; Bernkopf (1966) gives the historical account. Schmidt (1907) singled out the sequence space $\ell^2$ of square-summable sequences and extended to function spaces the orthogonalization procedure now called Gram–Schmidt, after him and after Gram. Riesz (1907) and Fischer (1907), within months of one another, proved that the square-integrable functions form a complete space isomorphic to $\ell^2$—a result resting on Lebesgue's (1902) integral, without which completeness of function spaces would be false. The abstract definition—an inner product space complete in norm—was named, and made the foundation of quantum mechanics, by von Neumann (1932/1955); Dirac (1958) supplied bra–ket notation and Stone (1932) the operator theory.

The language reaches far beyond quantum physics. Fourier analysis is Hilbert-space theory with a particular basis, and signal processing and numerical analysis lean on the same expansion theorems (Young, 1988). In quantum information the theory simplifies dramatically—a qubit is a unit vector in $\mathbb{C}^2$—while quantizing the electromagnetic field builds the infinite-dimensional Fock space from a countable collection of oscillators. One framework, from a two-level atom to a field.

Each idea below is stated once, proved in the simplest setting where the proof is transparent, and then interpreted physically, with worked examples doing the arithmetic that a general argument only sketches. Sections 2–4 build the geometry, the completeness that gives the space its name, and the expansion theory; Sections 5–7 turn to operators, the postulates of quantum mechanics, and the position and momentum bases.

## Vector Spaces and the Inner Product

Quantum mechanics begins with the observation that states add. A system that can be in state $A$ or state $B$ can equally be in a superposition $\alpha A + \beta B$ with complex coefficients; interference in the two-slit experiment is the clearest evidence that the scalars are complex, since probabilities add only *after* the underlying amplitudes have been added. The object supporting such superpositions is a vector space—a set $X$ closed under addition and scalar multiplication—in which a *basis* is a linearly independent spanning set, so that every element has a unique representation as a finite combination of basis vectors.

An *inner product* on $X$ is a map $\langle x, y\rangle$ from $X \times X$ to $\mathbb{C}$ such that for all $x, y, z \in X$ and $\alpha, \beta \in \mathbb{C}$,

$$\begin{aligned}
&\langle x, \alpha y + \beta z\rangle = \alpha\langle x, y\rangle + \beta\langle x, z\rangle, && \text{(linear in the second argument)},\\
&\langle x, y\rangle^* = \langle y, x\rangle, && \text{(conjugate symmetry)},\\
&\langle x, x\rangle \geq 0, \qquad \langle x, x\rangle = 0 \iff x = 0, && \text{(positive definiteness)}.
\end{aligned} \tag{1}$$

The inner product measures the *overlap* of two vectors. It is linear in its second argument and conjugate linear in the first, following the convention standard in physics. Two examples anchor the definition. On $\mathbb{C}^n$ the natural product is $\langle a, b\rangle = \sum_{i=1}^{n} a_i^* b_i$. On the space $C[a, b]$ of complex-valued continuous functions,

$$\langle f, g\rangle = \int_a^b f(x)^*\, g(x)\, dx, \tag{2}$$

the mean-square overlap of two signals. Both are vector spaces with a notion of overlap; Section 3 shows what separates them.

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**Key idea.** *The single most useful inequality in the subject follows from the axioms alone. For all $x, y$,*

$$|\langle x, y\rangle|^2 \le \langle x, x\rangle\, \langle y, y\rangle, \tag{3}$$

*with equality if and only if $x$ and $y$ are linearly dependent.*

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To see why, apply positive definiteness to $x - \lambda y$ with $\lambda = \langle y, x\rangle / \langle y, y\rangle$, the value minimizing its length. Expanding,

$$0 \le \|x - \lambda y\|^2 = \langle x, x\rangle - \frac{|\langle x, y\rangle|^2}{\langle y, y\rangle}, \tag{4}$$

and rearranging yields (3). Equality forces $x - \lambda y = 0$, so the vectors are dependent. This is the Cauchy–Schwarz inequality.

Because $\langle x, x\rangle \ge 0$, it makes sense to define the *norm* $\|x\| = \sqrt{\langle x, x\rangle}$, the length of $x$, and the *distance* $d(x, y) = \|x - y\|$. Cauchy–Schwarz is exactly what the norm needs in order to satisfy the triangle inequality,

$$\|x + y\| \le \|x\| + \|y\|, \tag{5}$$

and hence to make the vector space a metric space in which limits can be discussed.

Two vectors are *orthogonal*, written $x \perp y$, if $\langle x, y\rangle = 0$. Orthogonality generalizes perpendicularity and carries a Pythagorean theorem,

$$x \perp y \quad \Longrightarrow \quad \|x + y\|^2 = \|x\|^2 + \|y\|^2, \tag{6}$$

obtained by expanding the norm and discarding the cross term. More generally, for an orthonormal set $\{e_1, \dots, e_k\}$ (each $e_i$ of unit norm, mutually orthogonal),

$$Px = \sum_{i=1}^{k} \langle e_i, x\rangle\, e_i \tag{7}$$

is the *orthogonal projection* of $x$ onto the subspace those vectors span: the difference $x - Px$ is

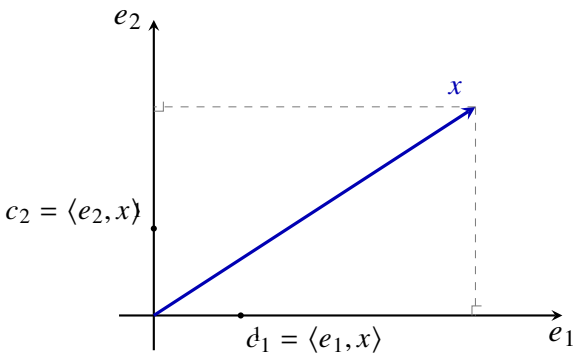



**Figure 1:** Orthogonal decomposition of $x$ in the orthonormal basis $\{e_1, e_2\}$. The components $c_i = \langle e_i, x \rangle$ are the projections of $x$ onto the basis directions, and $\|x\|^2 = |c_1|^2 + |c_2|^2$.

orthogonal to every $e_i$, and $Px$ is the point of that subspace closest to $x$. The components $c_i = \langle e_i, x \rangle$ are the shadows $x$ casts on the basis directions (Figure 1), and Pythagoras gives

$$\|x\|^2 = \sum_{i=1}^{k} |c_i|^2 + \|x - Px\|^2 \geq \sum_{i=1}^{k} |c_i|^2. \tag{8}$$

The projection formula is the geometry of measurement in miniature, and it returns in Section 6 as the Born rule.

**Worked example.** In $\mathbb{C}^2$ take $a = (1, i)$ and $b = (i, 1)$. Then $\langle a, b \rangle = 1^*(i) + (-i)(1) = 0$, so $a \perp b$, while $\langle a, a \rangle = \langle b, b \rangle = 2$. Both have norm $\sqrt{2}$, and $|\langle a, b \rangle|^2 = 0 < 4$, so Cauchy–Schwarz holds strictly—as it must, the vectors being independent. The equality case is realized instead by $x = (1, 0)$ and $y = (2, 0)$, where $|\langle x, y \rangle|^2 = 4 = \langle x, x \rangle \langle y, y \rangle$.

## Completeness: The Defining Property

Analysis in infinite dimensions is a story about limits, and the inner product supplies the topology they need. A sequence $x_n$ *converges* to $x$ in norm when $\|x_n - x\| \to 0$. A *Cauchy sequence* is one whose terms eventually agree to any prescribed precision: for every tolerance $\varepsilon$ there is an index $N$ such that $\|x_n - x_m\| < \varepsilon$ whenever $n, m \geq N$. Every convergent sequence is Cauchy, but the converse can fail—a sequence may be Cauchy yet converge to nothing inside the space.

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**Key idea.** *A **Hilbert space** is an inner product space that is* complete*: every Cauchy sequence converges to a limit belonging to the space itself.*

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Finite-dimensional spaces are complete for a dull reason: there a Cauchy sequence is just a list of componentwise Cauchy sequences of numbers, each of which converges. In infinite dimensions

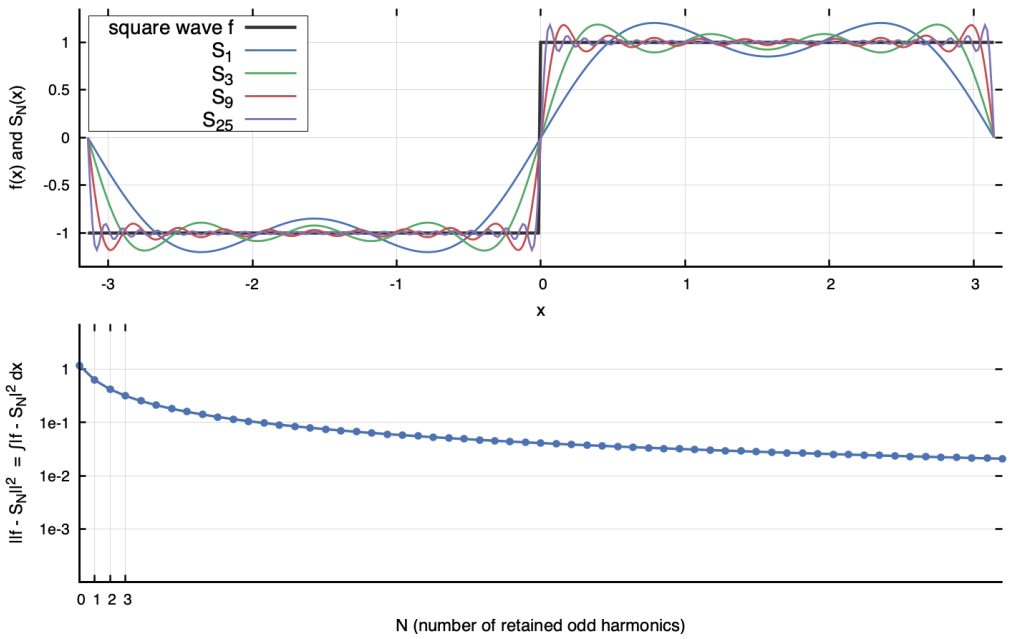



**Figure 2:** Completeness in pictures. Top: partial sums $S_N$ of the odd square wave $f$ on $[-\pi, \pi]$, for $N = 1, 3, 9, 25$. Each $S_N$ is continuous and the sequence approaches $f$ in the mean, while overshooting at the jumps. Bottom: the mean-square error $\|f - S_N\|^2$ versus the number of retained odd harmonics on a logarithmic scale, decaying to zero. The approximation converges in $L^2[-\pi, \pi]$ although the limit function is discontinuous—precisely why the continuous functions fail to form a Hilbert space and $L^2$ succeeds.

completeness is a real constraint. The canonical failure is $C[-1, 1]$, the continuous functions with the mean-square norm. The ramp functions

$$f_n(x) = \begin{cases} -1, & x \le -1/n, \\ nx, & -1/n \le x \le 1/n, \\ 1, & x \ge 1/n, \end{cases} \qquad n = 1, 2, \ldots \tag{9}$$

are each continuous, and $\{f_n\}$ is Cauchy in the mean-square metric, since the functions differ only near $x = 0$, over an interval that shrinks as $n$ grows. Yet the pointwise limit is the square wave $\operatorname{sgn}(x)$, which jumps at the origin and is *not* continuous, so the sequence has no limit in $C[-1, 1]$.

The same phenomenon appears in Fourier series (Figure 2): the partial sums $S_N$ of the square wave are continuous and approximate it ever more closely in the mean, but the limiting function is discontinuous. The approximation converges; the question is whether the limit is allowed to live in the space. In $C[-\pi, \pi]$ it is not.

The repair is to enlarge the notion of function, and the tool is the integral. Lebesgue (1902) rebuilt integration from measure theory so that mean-square limits of reasonable functions are again integrable; the square wave is Lebesgue integrable, with $\int \operatorname{sgn}^2 = 2\pi$. The space $L^2[a, b]$ consists

of the measurable functions $f$ with finite norm

$$\|f\| = \left( \int_a^b |f(x)|^2 \, dx \right)^{1/2}, \tag{10}$$

with two functions identified when they differ only on a set of measure zero—the identification that makes $\|f\| = 0$ imply $f = 0$, as axiom (1) requires. Fischer (1907) proved that every Cauchy sequence in $L^2[a, b]$ then converges to an element of the space, and Riesz (1907) the corresponding statement for the coefficients of expansions. Together they are the Riesz–Fischer theorem, and completeness is restored.

The canonical Hilbert spaces are therefore $\mathbb{C}^n$; Schmidt's (1907) sequence space

$$\ell^2 = \left\{ (a_1, a_2, \dots) : a_i \in \mathbb{C},\ \sum_{i=1}^{\infty} |a_i|^2 < \infty \right\}, \qquad \langle a, b \rangle = \sum_{i=1}^{\infty} a_i^* b_i, \tag{11}$$

the function space $L^2[a, b]$; and $L^2(\mathbb{R}^3)$, the state space of a spinless particle, where $\int |\psi|^2$ is the total probability. All are *separable*—each contains a countable dense subset—which guarantees a countable basis, as Section 4 requires.

Why does completeness matter physically? Quantum states are routinely defined by infinite sums; the expansion $x = \sum c_n e_n$ is honest only if its partial sums converge to an element of the space, and completeness is precisely the assurance that they do. In $C[-\pi, \pi]$ the square wave is not a state at all, though it is the limit of states $S_N$; in $L^2$ it is.

## Orthonormal Bases and Expansions

An *orthonormal system* is a set of mutually orthogonal unit vectors, $\langle e_i, e_j \rangle = \delta_{ij}$. It is *complete*, or an *orthonormal basis*, if it cannot be enlarged—equivalently, if no nonzero vector is orthogonal to every $e_i$, or again if its finite linear combinations are dense. In an incomplete inner product space such a basis need not exist; in a separable Hilbert space it always does.

The *Bessel inequality* extends the finite-dimensional bound of Section 2 to any orthonormal system:

$$\sum_i |\langle e_i, x \rangle|^2 \le \|x\|^2. \tag{12}$$

Its content is that the coordinate sequence of $x$ is square-summable for *every* orthonormal system; it is proved by applying (6) to the projections $x - P_N x$ and letting $N \to \infty$. This is why the coefficients of a state are well behaved: the squared amplitudes of a state sum to at most one.

Existence of bases is constructive. Given a countable linearly independent set, the *Gram–Schmidt procedure* builds an orthonormal set spanning the same subspace by successively subtracting the

components already accounted for:

$$e_1 = \frac{v_1}{\|v_1\|}, \qquad e_n = \frac{v_n - \sum_{i=1}^{n-1} \langle e_i, v_n \rangle e_i}{\|v_n - \sum_{i=1}^{n-1} \langle e_i, v_n \rangle e_i\|}. \tag{13}$$

This is Schmidt's (1907) construction; applied to a countable dense subset of a separable Hilbert space it yields a countable orthonormal basis.

**Worked example.** Orthogonalize $v_1 = (1, 1)$ and $v_2 = (1, 0)$ in $\mathbb{R}^2$. First $e_1 = v_1/\|v_1\| = (1, 1)/\sqrt{2}$. Since $\langle e_1, v_2 \rangle = 1/\sqrt{2}$,

$$v_2 - \langle e_1, v_2 \rangle e_1 = (1, 0) - \tfrac{1}{2}(1, 1) = \left(\tfrac{1}{2}, -\tfrac{1}{2}\right),$$

of norm $1/\sqrt{2}$; hence $e_2 = (1, -1)/\sqrt{2}$. The result is the familiar rotated pair, and indeed $\langle e_1, e_2 \rangle = 0$ with both of unit length.

If $\{e_n\}$ is an orthonormal basis, every vector expands as

$$x = \sum_{n=1}^{\infty} c_n e_n, \qquad c_n = \langle e_n, x \rangle, \tag{14}$$

with convergence in norm, and the coefficients determine the vector completely: $\langle e_n, x \rangle = 0$ for all $n$ forces $x = 0$. The norm is recovered from the coefficients by *Parseval's identity*,

$$\|x\|^2 = \sum_{n=1}^{\infty} |c_n|^2, \tag{15}$$

the infinite-dimensional Pythagorean theorem, obtained by expanding the norm and using continuity of the inner product. It is why a total probability equals the sum of the probabilities of the individual outcomes.

The map $x \mapsto (c_n)$ is linear, preserves the inner product, and has range all of $\ell^2$: this is the *Riesz–Fischer theorem*, whose halves are due to Riesz (1907) and Fischer (1907). Given any square-summable sequence $(c_n)$, the partial sums $\sum_{n=1}^{N} c_n e_n$ form a Cauchy sequence, so by completeness they converge to some $x$ having those coefficients. Thus $L^2[a, b]$ and $\ell^2$ are one space in two descriptions, and since lengths are preserved by (15) the isomorphism is *isometric*.

The archetype is Fourier analysis. The functions

$$\frac{1}{\sqrt{2\pi}}, \quad \frac{\cos nx}{\sqrt{\pi}}, \quad \frac{\sin nx}{\sqrt{\pi}} \qquad (n = 1, 2, \dots) \tag{16}$$

form an orthonormal basis of $L^2[-\pi, \pi]$, and (14) is the classical Fourier series; that the system is complete is a theorem about $L^2$, not an assumption. The square wave of Section 3 is the simplest

nontrivial case, with coefficients

$$c_{2k} = 0, \qquad c_{2k+1} = \frac{4}{\sqrt{\pi}(2k+1)}, \tag{17}$$

so the series is the familiar odd-harmonic sine series whose partial sums appear in Figure 2.

**Worked example.** Parseval's identity can be checked against a known integral. Here $\|f\|^2 = \int_{-\pi}^{\pi} \operatorname{sgn}(x)^2\, dx = 2\pi$, while

$$\sum_{k=0}^{\infty} c_{2k+1}^2 = \frac{16}{\pi} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{16}{\pi} \cdot \frac{\pi^2}{8} = 2\pi,$$

using $\sum_{\text{odd}} k^{-2} = \pi^2/8$. The identity holds even though the target function is discontinuous: the overshoot in Figure 2 is a pointwise phenomenon that leaves convergence in the mean untouched.

## Linear Operators and the Spectral Theorem

Measurement is an operation on states, and the mathematical image of an operation is an operator. A *linear operator* on a Hilbert space $\mathcal{H}$ is a linear map $A$ from $\mathcal{H}$, or a dense subspace of it, into itself. It is *bounded* if

$$\|A\| = \sup_{\|x\|=1} \|Ax\| < \infty, \tag{18}$$

and boundedness is equivalent to continuity, the property that $x_n \to x$ implies $Ax_n \to Ax$. Bounded operators are the matrices of infinite dimension; the operators of quantum mechanics—position, momentum, and the Hamiltonian—are unbounded.

The *adjoint* $A^\dagger$ of a bounded operator $A$ is the unique operator satisfying

$$\langle x, Ay \rangle = \langle A^\dagger x, y \rangle \qquad \text{for all } x, y \in \mathcal{H}, \tag{19}$$

the rule that moves an operator across a bracket to the other side, with conjugation. It is the infinite-dimensional analogue of the conjugate transpose of a matrix. An operator is *self-adjoint* (Hermitian) if $A^\dagger = A$.

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**Key idea.** *For a self-adjoint operator the eigenvalues are real, and eigenvectors belonging to distinct eigenvalues are orthogonal:*

$$Ax = \lambda x, \quad Ay = \mu y, \quad \lambda \neq \mu \quad \Longrightarrow \quad \langle x, y \rangle = 0. \tag{20}$$

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Both facts are one-line bracket manipulations. Reality: $\lambda\langle x, x\rangle = \langle x, Ax\rangle = \langle Ax, x\rangle = \lambda^*\langle x, x\rangle$. Orthogonality: $\lambda\langle x, y\rangle = \langle Ax, y\rangle = \langle x, Ay\rangle = \mu\langle x, y\rangle$, so $(\lambda - \mu)\langle x, y\rangle = 0$. Eigenvalues are the possible outcomes of a measurement and must be real numbers; self-adjointness is what guarantees it.

For unbounded operators a subtlety enters, clarified by von Neumann (1932/1955). Position and momentum act only on a dense subspace, since not every square-integrable function is differentiable. Being *symmetric*—satisfying $\langle x, Ay\rangle = \langle Ax, y\rangle$ on a common domain—is weaker than being *self-adjoint*, which requires that the adjoint have the same domain. A symmetric operator that is not self-adjoint has no spectral theorem and generates no unitary evolution; von Neumann's deficiency-index theory decides when promotion is possible, and hence whether the Schrödinger equation has a solution for all time.

The *spectral theorem* states that a self-adjoint operator can be diagonalized. In its discrete form, the eigenvectors of a self-adjoint operator with pure point spectrum form an orthonormal basis, and the operator acts by multiplication by its eigenvalues; the continuous spectrum, where eigenvectors are not square-integrable, is handled by the generalized version of Section 7. Physically the theorem says that a measurement decomposes the state space into eigenspaces, so that a state is fixed by a complete set of commuting observables (Steen, 1973).

## Quantum Mechanics in Hilbert Space

The postulates of quantum mechanics are statements about a Hilbert space, first put in this form by von Neumann (1932/1955).

**States.**
A physical system is described by a unit vector $|\psi\rangle$ in a separable Hilbert space $\mathcal{H}$, determined only up to a phase—a ray.

**Observables.**
A measurable quantity is a self-adjoint operator on $\mathcal{H}$.

**Measurement.**
The possible outcomes are the eigenvalues of the operator. If the spectrum is discrete and $e_\lambda$ is the normalized eigenvector for eigenvalue $\lambda$, the probability of outcome $\lambda$ in state $|\psi\rangle$ is

$$P(\lambda) = |\langle e_\lambda, \psi\rangle|^2, \tag{21}$$

the Born rule.

**Evolution.**
Between measurements the state evolves unitarily, generated by the Hamiltonian through the

**Table 1:** The dictionary between Hilbert-space geometry and quantum mechanics.

| Hilbert space | Quantum mechanics |
|---|---|
| Unit vector $\lvert\psi\rangle$ (a ray) | State of the system |
| Inner product $\langle\varphi\lvert\psi\rangle$ | Probability amplitude |
| Orthonormal basis $\{\lvert n\rangle\}$ | Complete set of measurement outcomes |
| Self-adjoint operator $A$ | Observable |
| Eigenvalue $\lambda$ of $A$ | Possible measured value |
| Projection $\lvert\langle n\lvert\psi\rangle\rvert^2$ | Probability of outcome $n$ (Born rule) |
| Unitary $e^{-iHt/\hbar}$ | Time evolution |
| Tensor product $\mathcal{H}_1 \otimes \mathcal{H}_2$ | Composite system |

Schrödinger equation

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle, \tag{22}$$

whose solution $|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$ exists for all times precisely because $H$ is self-adjoint.

Dirac's (1958) bra–ket notation makes these postulates fluent. A vector is a *ket* $|\psi\rangle$; the linear functional $\langle\varphi, \cdot\,\rangle$ is the *bra* $\langle\varphi|$; the pairing is the inner product $\langle\varphi|\psi\rangle = \langle\varphi, \psi\rangle$; and the rank-one projection onto the direction of $|\psi\rangle$ is written $|\psi\rangle\langle\psi|$. For an orthonormal basis $\{|n\rangle\}$, the projection formula (7) becomes the *resolution of the identity*,

$$\sum_n |n\rangle\langle n| = 1, \tag{23}$$

an equality of operators; applied to $|\psi\rangle$ it recovers the expansion (14). Compare this with (21): the component of a vector along a basis direction is an inner product, and the probability of finding the state there is the squared modulus of that same number. Measurement is projection—the geometry of Section 2, read as probability. Table 1 collects the correspondence.

**Worked example.** Take a spin-$\frac{1}{2}$ particle in the state $|\psi\rangle = |\uparrow_z\rangle$ and measure $S_x$, whose eigenstates are $|\pm_x\rangle = (|\uparrow_z\rangle \pm |\downarrow_z\rangle)/\sqrt{2}$ with eigenvalues $\pm\hbar/2$. The Born rule gives

$$P(+) = |\langle +_x|\uparrow_z\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2},$$

and likewise $P(-) = \frac{1}{2}$: the amplitudes have equal modulus, so the two outcomes are equally likely even though the state was definite before the measurement. Here $\mathcal{H} = \mathbb{C}^2$, complete trivially because it is finite dimensional—the same space, under another name, as a qubit.

Superposition and interference follow from linearity. If $|\psi\rangle = \alpha|a\rangle + \beta|b\rangle$ with $\alpha, \beta \in \mathbb{C}$, then

$$P(\psi \to c) = |\alpha\langle c, a\rangle + \beta\langle c, b\rangle|^2 = |\alpha|^2 P(a \to c) + |\beta|^2 P(b \to c) + 2\,\mathrm{Re}\big(\alpha^*\beta\,\langle a, c\rangle^*\langle b, c\rangle\big), \tag{24}$$

and the cross term is the interference that produces two-slit fringes. The relative phase of $\alpha$ and $\beta$

is physical and cannot be removed by a change of basis. That states are vectors, and probabilities squared inner products, is the entire origin of interference.

Entanglement, finally, is a statement about tensor products: two spins live in $\mathbb{C}^2 \otimes \mathbb{C}^2$, a four-dimensional space containing states such as $(|{\uparrow\downarrow}\rangle - |{\downarrow\uparrow}\rangle)/\sqrt{2}$ that factor into no product of single-particle states. Superposition, measurement, uncertainty, and entanglement all appear in this miniature; the infinite-dimensional spaces enter with the continuum, to which the next section turns.

## The Position and Momentum Representations

A wave function is not a different kind of object from a state; it is that state read in a particular basis. In the *position representation* the basis states are the localized kets $|x\rangle$, and the wave function is the component $\psi(x) = \langle x|\psi\rangle$, the amplitude to find the particle at $x$. In the *momentum representation* it is $\widetilde{\psi}(p) = \langle p|\psi\rangle$. The two are related by the overlap

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\, e^{ipx/\hbar}, \tag{25}$$

a unitary change of basis; substituting the resolution of the identity turns it into the Fourier transform

$$\widetilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi(x)\, e^{-ipx/\hbar}\, dx, \qquad \psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \widetilde{\psi}(p)\, e^{ipx/\hbar}\, dp. \tag{26}$$

Plancherel (1910) proved that this transform is an isometric isomorphism of $L^2(\mathbb{R})$ onto itself, which is Parseval's identity (15) in continuous form:

$$\int |\psi(x)|^2\, dx = \int |\widetilde{\psi}(p)|^2\, dp. \tag{27}$$

The constant $1/\sqrt{2\pi\hbar}$ is fixed by requiring that both wave functions be normalized in the same way.

The subtlety is that the kets $|x\rangle$ and $|p\rangle$ are not themselves members of the Hilbert space: they are delta-normalized, $\langle x|x'\rangle = \delta(x - x')$, and $\langle x, x\rangle$ is not finite. They are *generalized eigenvectors* of the position and momentum operators, which in this representation act as multiplication by $x$ and as $-i\hbar\,\partial/\partial x$. The framework that contains them is the *rigged Hilbert space*, or Gelfand triple,

$$\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R}) \subset \mathcal{S}'(\mathbb{R}), \tag{28}$$

where $\mathcal{S}$ is the Schwartz space of smooth rapidly decreasing functions and $\mathcal{S}'$ its dual space of tempered distributions (Prugovečki, 1981). The square-integrable functions, the legitimate states, sit in the middle; the localized kets and plane waves lie outside the Hilbert space but inside the distributions, where they act as continuous linear functionals (Figure 3). Dirac notation is, in this setting, a well-defined shorthand for that structure.

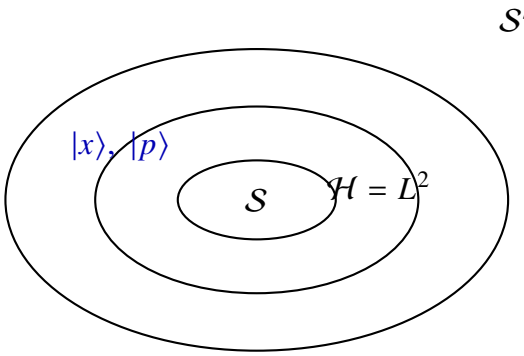



**Figure 3:** The rigged Hilbert space (Gelfand triple) $\mathcal{S} \subset L^2 \subset \mathcal{S}'$. The Dirac kets $|x\rangle$ and $|p\rangle$ are not square-integrable and lie outside the Hilbert space $L^2$, but they are legitimate elements of the distribution space $\mathcal{S}'$.

The position and momentum operators obey the canonical commutation relation $[X, P] = i\hbar$, and the *Stone–von Neumann theorem* states that any irreducible pair of self-adjoint operators satisfying it is unitarily equivalent to the position and momentum operators on $L^2(\mathbb{R})$. Proved by Stone (1932) and von Neumann (1932/1955), the theorem is why wave mechanics and matrix mechanics are one theory rather than two. From it follows the *uncertainty relation*

$$\Delta x\, \Delta p \geq \frac{\hbar}{2}, \qquad \Delta A = \sqrt{\langle \psi, A^2\psi\rangle - \langle \psi, A\psi\rangle^2}, \tag{29}$$

where $\Delta A$ is the standard deviation of $A$ in the state. The proof is a two-line application of Cauchy–Schwarz (3) to the vectors $(X - \langle\psi, X\psi\rangle)\psi$ and $(P - \langle\psi, P\psi\rangle)\psi$: the uncertainty relation is a statement about Hilbert-space geometry, not about the disturbance caused by measurement. A state sharply localized in position must be spread out in momentum, the two profiles being related by the Fourier transform of (26).

## Conclusion

From a single idea, the inner product, the whole structure has unfolded. The inner product supplied the geometry—norms, orthogonality, Cauchy–Schwarz, projection. Completeness made that geometry analytic: in $L^2$ the mean-square limits that fail among continuous functions become proper elements of the space. Orthonormal bases turned vectors into sequences of numbers and fused $L^2$ with $\ell^2$ through the Riesz–Fischer theorem, and self-adjoint operators with the spectral theorem supplied the language of observables. Assembled, these pieces are quantum mechanics: states are unit vectors, observables are self-adjoint operators, probabilities are squared inner products, and evolution is unitary. The position and momentum representations show one state in two bases connected by a Fourier transform, and the rigged Hilbert space shows where the delta-normalized kets of Dirac notation genuinely live.

The same structure serves at every scale: the two-slit experiment, the energy levels of hydrogen, the two levels of a qubit, and the infinite-dimensional field of quantum electrodynamics are one mathematics at different dimensions. What makes it honest rather than formal is completeness—the infinite sums of quantum mechanics are genuine limits of states. For the mathematics, see Halmos (1951), Reed and Simon (1980), and Kreyszig (1978), with Riesz and Sz.-Nagy (1955) as the classic connecting text; for the physics, von Neumann (1932/1955), Dirac (1958), Sakurai and Napolitano (2020), and Griffiths and Schroeter (2018).

## References


Bernkopf, M. (1966). The development of function spaces with particular reference to their origins in integral equation theory. *Archive for History of Exact Sciences, 3*(1), 1–96. https://doi.org/10.1007/BF00412288

Dirac, P. A. M. (1958). *The principles of quantum mechanics* (4th ed.). Oxford University Press.

Fischer, E. (1907). Sur la convergence en moyenne. *Comptes rendus hebdomadaires des séances de l'Académie des sciences, 144*, 1022–1024.

Griffiths, D. J., & Schroeter, D. F. (2018). *Introduction to quantum mechanics* (3rd ed.). Cambridge University Press.

Halmos, P. R. (1951). *Introduction to Hilbert space and the theory of spectral multiplicity*. Chelsea.

Hilbert, D. (1912). *Grundzüge einer allgemeinen Theorie der linearen Integralgleichungen*. B. G. Teubner.

Kreyszig, E. (1978). *Introductory functional analysis with applications*. John Wiley & Sons.

Lebesgue, H. (1902). Intégrale, longueur, aire. *Annali di Matematica Pura ed Applicata, 7*, 231–359. https://doi.org/10.1007/BF02420592

Plancherel, M. (1910). Contribution à l'étude de la représentation d'une fonction arbitraire par les intégrales définies. *Rendiconti del Circolo Matematico di Palermo, 30*, 289–335.

Prugovečki, E. (1981). *Quantum mechanics in Hilbert space* (2nd ed.). Academic Press.

Reed, M., & Simon, B. (1980). *Methods of modern mathematical physics: Vol. 1. Functional analysis* (Rev. ed.). Academic Press.

Riesz, F. (1907). Sur les systèmes orthogonaux de fonctions. *Comptes rendus hebdomadaires des séances de l'Académie des sciences, 144*, 615–619.

Riesz, F., & Sz.-Nagy, B. (1955). *Functional analysis*. Ungar.

Sakurai, J. J., & Napolitano, J. (2020). *Modern quantum mechanics* (3rd ed.). Cambridge University Press.

Schmidt, E. (1907). Zur Theorie der linearen und nichtlinearen Integralgleichungen. I. Teil: Entwicklung willkürlicher Funktionen nach Systemen vorgeschriebener. *Mathematische Annalen, 63*, 433–476.

Steen, L. A. (1973). Highlights in the history of spectral theory. *The American Mathematical Monthly, 80*(4), 359–381. https://doi.org/10.1080/00029890.1973.11993292

Stone, M. H. (1932). *Linear transformations in Hilbert space and their applications to analysis* (AMS Colloquium Publications, Vol. 15). American Mathematical Society.

von Neumann, J. (1955). *Mathematical foundations of quantum mechanics*. Princeton University Press. (Original work published 1932)

Young, N. (1988). *An introduction to Hilbert space*. Cambridge University Press.